

# Effectiveness of Sitagliptin and Metformin FDC in Indian Patients with T2DM from a Real-World Retrospective EMR Based Study: A Subgroup Analysis in Treatment Naive Patients

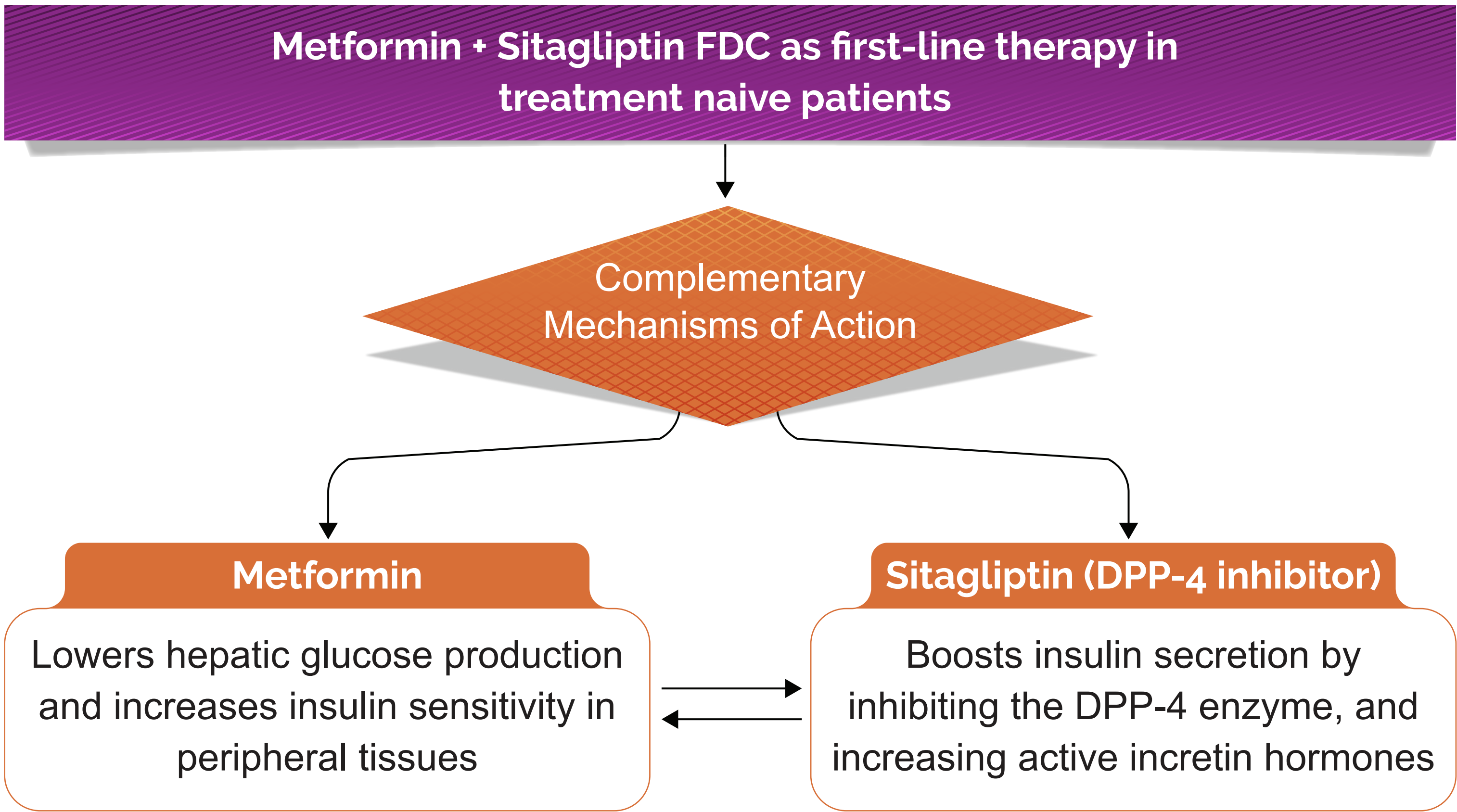
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## BACKGROUND

- According to the 2021 ICMR-INDIAB-17 national cross-sectional study, diabetes prevalence in India reached 11.4%, with an additional 15.3% of the population diagnosed with prediabetes.<sup>1</sup>
- Managing type 2 diabetes mellitus (T2DM) in India presents unique challenges, including early disease onset<sup>2</sup>, high rates of complications<sup>3,4</sup>, and a growing population of treatment-naive patients who require effective initial therapies to achieve glycemic control and prevent progression.
- Due to the progressive nature of T2DM, guidelines recommend a two- or three-drug regimen based on HbA1c levels.<sup>5,6</sup> Combination therapy, especially fixed-dose combinations (FDCs) simplify treatment, enhances adherence, and offers superior glycemic control over monotherapy.<sup>6</sup>
- Sitagliptin and Metformin FDC is widely used to effectively manage T2DM due to their complementary mechanisms of action.

Figure 1: Metformin and sitagliptin: Complementary mechanisms of action<sup>7</sup>



- Real-world evidence, particularly from retrospective electronic medical records (EMR)-based studies, provides a comprehensive view of treatment outcomes across diverse patient populations.<sup>8</sup>
- The current study evaluated the practical real-world insights of the sitagliptin-metformin FDC use among treatment-naive T2DM patients.

## AIM/ OBJECTIVE

- This real-world, retrospective observational study using EMR data was conducted to evaluate the efficacy of the sitagliptin + metformin FDC in Indian patients with type 2 diabetes mellitus (T2DM).
- This subgroup analysis presented data for treatment-naive patients with baseline HbA1c levels of  $\geq 7.5\%$ .

## METHODOLOGY

### Study design

- This real-world, retrospective, observational, EMR based study gathered data from January 2017 to March 2023 from EMRs of 206,335 patients who were prescribed with sitagliptin + metformin FDC alone or with other oral anti-diabetic medications (Figure 2).
- Study was approved by Independent Ethics Committee and registered on CTRI (CTRI no: CTRI/2023/10/058366)

### Inclusion criteria

- Male and female patients with type-2 diabetes mellitus  $\geq 18$  years of age.
- Patients prescribed with sitagliptin alone or with other oral antidiabetic medications OR patients prescribed with sitagliptin + metformin FDC alone or with other oral anti-diabetic medications.
- Patients with EMR data captured for more than one visit.

### Exclusion criteria

- Patients with type-1 diabetes mellitus.
- Patients on insulin or any other injectable antidiabetic medication like glucagon like peptide-1 (GLP -1) agonists.
- Patients with incomplete medical records with respect to the required study parameters.

### Primary Endpoint

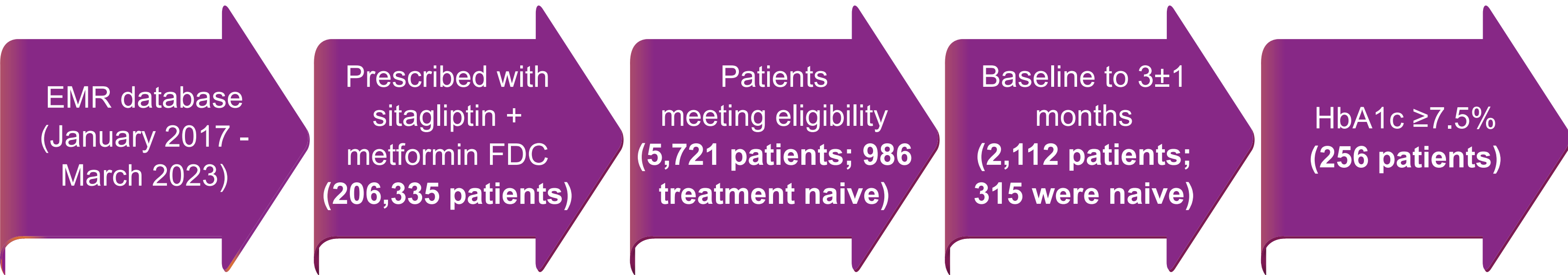
Mean change in HbA1c in patients with baseline HbA1c  $\geq 7\%$  in sitagliptin + metformin FDC arm from baseline to 3 months.

### Key Secondary Endpoints

- Mean change in HbA1c from baseline to 3 months in treatment-naive patients with baseline HbA1c  $\geq 7.5\%$  in sitagliptin and metformin FDC arm.
- Mean change in fasting blood glucose (FBG), and postprandial blood glucose (PPBG) from baseline to 3 months in treatment-naive patients with baseline HbA1c  $\geq 7\%$  in sitagliptin and metformin FDC arm.
- Proportion of patients achieving HbA1c  $< 7\%$  in sitagliptin and metformin FDC arm.

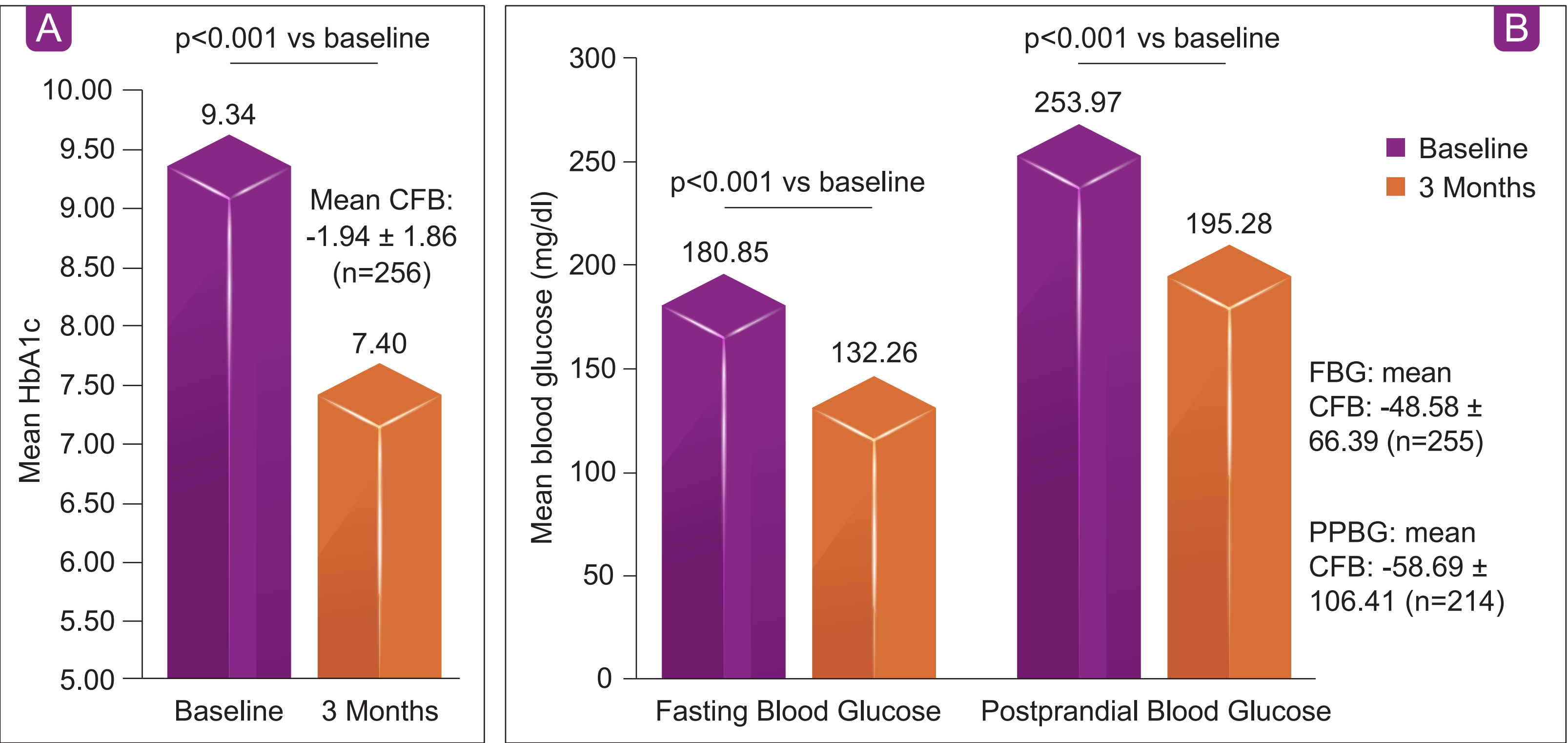
## RESULTS

Figure 2: Patient flow



- 5,721 patients met the eligibility criteria (HbA1c  $\geq 7\%$ ; prescribed with FDC in any visit other than the baseline visit on EMR platform and had visits as per the study time points) and 986 of them were treatment naive (Figure 2).
- Results for primary endpoint (n=315): Mean HbA1c reduced from  $8.94 \pm 1.78\%$  at baseline to  $7.33 \pm 1.05\%$  at 3 months.

Figure 3: Mean changes in (A) HbA1c, (B) FBG and PPBG from baseline to 3 months



- A significant reduction in HbA1c levels was observed over 3 months ( $9.34 \pm 1.74$  to  $7.40 \pm 1.11$ ; CFB:  $-1.94 \pm 1.86$ , p<0.001) (Figure 3).
- Both FBG ( $180.85 \pm 59.19$  to  $132.26 \pm 39.58$ ; CFB:  $-48.58 \pm 66.39$ , p<0.001) and PPBG ( $253.97 \pm 84.87$  to  $195.28 \pm 71.64$ ; CFB:  $-58.69 \pm 106.41$ , p<0.001) levels significantly decreased from baseline (Figure 3).
- Proportion of patients achieving HbA1c  $< 7\%$  at 3 months: 38.7% (n=99/256).
- **Limitation:** This was a single arm analysis of EMR data captured on digital platform, and the database that we used does not capture safety data.

## CONCLUSION

- This subgroup analysis highlighted the effectiveness of sitagliptin and metformin FDC in improving HbA1c in treatment-naive T2DM patients with baseline HbA1c  $\geq 7.5\%$ .
- Use of FDC also demonstrated significant improvements in FBG and PPBG levels in this subgroup of patients.
- These results emphasize the role of sitagliptin and metformin FDC as valuable therapeutic option in managing hyperglycemia in this subset of population.

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